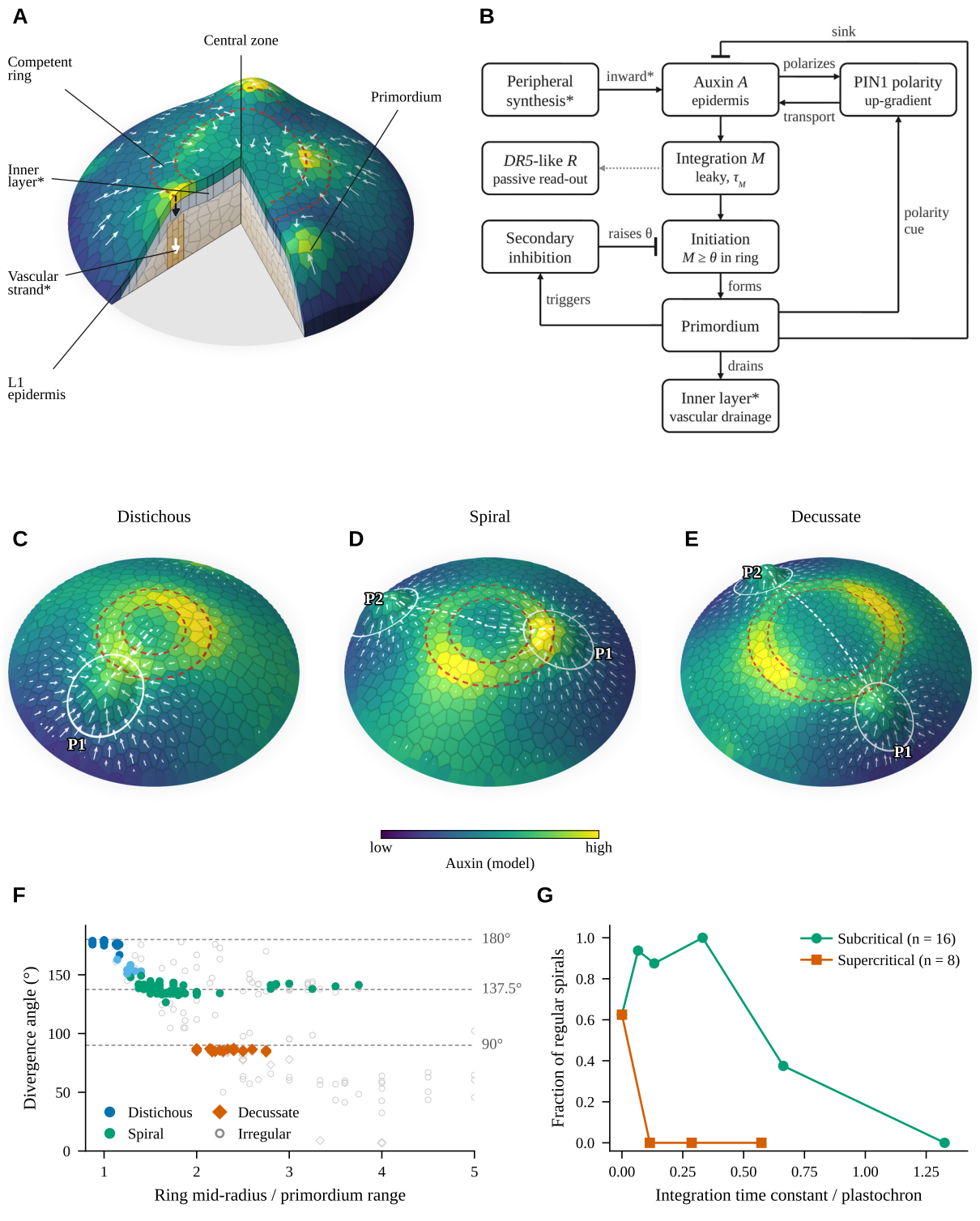

A
Central zone
Competent ring
Primordium
Inner layer*
Vascular strand*
L1 epidermis
B
sink
Peripheral synthesis*
inward*
Auxin A epidermis
polarizes
PIN1 polarity up-gradient
transport
DR5-like R passive read-out
Integration M leaky, τM
polarity cue
Secondary inhibition
raises θ
Initiation M ≥ θ in ring
forms
triggers
Primordium
drains
Inner layer* vascular drainage
Distichous
Spiral
Decussate
C
D
E
P1
P2
low
high
Auxin (model)
F
G
Divergence angle (°)
180°
137.5°
90°
Distichous
Spiral
Decussate
Irregular
Ring mid-radius / primordium range
Fraction of regular spirals
Subcritical (n = 16)
Supercritical (n = 8)
Integration time constant / plastochron